

Grading criteria

Part 1: Speech Recording and Data Preparation (20%)

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Speech Recording	All 11 recordings present, clearly pronounced, mono format, 44.1 kHz, correct duration and clean audio. Correct naming convention	Minor issue in 1 recording (slight noise or formatting)	Several recordings contain noise or small formatting issues. Some naming inconsistencies or misplaced files	Multiple recordings missing or incorrect format. Files largely incorrectly named or missing	10%
Data Loading and DataFrame Construction	Correctly loads recordings and constructs DataFrame with correct columns	Minor coding issue but DataFrame correctly constructed	DataFrame created but partially incorrect (missing some entries or metadata)	DataFrame incorrectly structured or incomplete	10%

Part 2: Formant Extraction with AI Assistance (20%)

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Correct Extraction of F1 and F2	Accurate formant extraction for all recordings using appropriate library	Minor implementation issues but most values correct	Some incorrect formant calculations or missing values	Major errors or incorrect approach	5%
AI Documentation	Identifies AI tool used. Exact prompts and responses clearly documented and explained. Clear explanation of signal-processing methods and libraries used. Strong explanation linking formant extraction to spectral analysis, FFT, vocal tract resonance. Clearly explains modifications made to AI-generated code and why	AI tool identified. Prompts and outcomes listed. Methods described but lacking technical detail. Some conceptual connection provided. Minor explanation of modifications	AI tool identified. Prompts and outcomes not listed. Partial explanations for methods. Weak conceptual connection. Limited explanation of modifications.	Not documented. Missing prompts. No method description. No connection to lecture. No modification explanations.	15%

Part 3: Formant Space Visualization (20%)

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Scatter Plot Generation	Plot correctly implemented axes, clear labels, legend, and readable visualization. Distinct colors for each speaker group and correct word annotations	Plot correct but minor formatting issues. Minor annotation or color inconsistencies	Plot generated but missing labels or formatting. Partial annotations	Plot incorrect or incomplete. Missing annotations	10%
Plot analysis	Insightful analysis of vowel clustering and acoustic similarity	Reasonable interpretation with minor limitations	Basic description without deep analysis	Incorrect or missing explanation	10%

Part 4: Pitch Estimation and Spectrogram Analysis (20%)

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Concatenation of Audio Files	Correct concatenation of recordings into three combined wav files			Incorrect or missing concatenation	5%
Pitch Estimation + visualization	Correct implementation of pyin. Clear spectrogram with F0 contour correctly overlaid	Visualization correct but minor formatting issues	Visualization present but unclear	Incorrect implementation	15%

Code Readability, Organization, and Documentation (20%)

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Marks
Code Readability	Clean, readable code with appropriate variable names and formatting	Mostly readable code	Some confusing or inconsistent code	Difficult to read	10%
Reproducibility and Execution	Notebook runs end-to-end without errors and produces expected outputs	Minor runtime issue	Requires manual fixes to run	Does not run	10%